Multivariate Calculus Lecture Notes
Section 17.3 – The Divergence Theorem

I. Idea

a. Recall the flux-divergence form of Green's Theorem:

$$\oint_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \operatorname{div} \mathbf{F}\, dA,$$

where ∂D is the boundary of the region D.

i. This says that the outward flux across the boundary of a region in the plane equals the integral of the divergence over that region.

ii. There's no reason why this should be restricted to 2D, so let's generalize it to solids in 3D.

iii. Instead of a double integral of the divergence over some 2D plane region D we would have a triple integral of the divergence over some 3D solid region W.

iv. And instead of a flux line integral around the closed boundary of D we would have a flux surface integral over the closed surface of W.

b. Divergence Theorem: Let S be a closed surface enclosing a region W in $\mathbb{R}^3$. Suppose S is piecewise smooth and oriented with normal vectors pointing outside of W. If $\mathbf{F}$ is a vector field whose domain contains W, then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_W \operatorname{div} \mathbf{F}\, dV.$$

II. Examples

a. Find the flux of $\mathbf{F} = -xz\mathbf{i} - yz\mathbf{j} + z^2\mathbf{k}$ over the surface S given by the ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$.

i. We have $\operatorname{div} \mathbf{F} = -z - z + 2z = 0$.

ii. Thus

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_W \operatorname{div} \mathbf{F}\, dV = \iiint_W 0\, dV = 0.$$

iii. It would take much more work to arrive at this answer using the surface integral.

b. Find the outward flux of $\mathbf{F} = z^2x\mathbf{i} + (\frac{y^3}{3} + \tan z)\mathbf{j} + (x^2z + y^2)\mathbf{k}$ across the surface consisting of the top half of a sphere of radius 1 centered at the origin and the disk of radius 1 centered at the origin.

i. Calculating this directly would require two surface integrals.

ii. However, $\operatorname{div} \mathbf{F} = z^2 + y^2 + x^2$, which isn't too bad.

iii. The flux across the solid S is given by

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_W (x^2 + y^2 + z^2)\, dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= 2\pi \left[-\cos\phi\right]_0^{\pi/2} \left[\frac{\rho^5}{5}\right]_0^1 = \frac{2\pi}{5}(1) = \frac{2\pi}{5}.$$

III. Some perspective

a. There are several applications of Stokes's Theorem and the Divergence Theorem to electricity and magnetism; see the text.
b. Major integral theorems
   i. Fundamental Theorem of Calculus: $\int_a^b f'(x)\,dx = f(b) - f(a)$.
   ii. Fundamental Theorem of Line Integrals: $\int_C \nabla f \cdot d\mathbf{r} = f(Q) - f(P)$, where C is a curve from P to Q.
   iii. Green's Theorem (circulation-curl form): $\iint\limits_D \operatorname{curl}_z \mathbf{F}\,dA = \oint_C \mathbf{F} \cdot d\mathbf{r}$.
   iv. Green's Theorem (flux-divergence form): $\iint\limits_D \operatorname{div} \mathbf{F}\,dA = \oint_C \mathbf{F} \cdot \mathbf{n}\,ds$.
   v. Stokes's Theorem: $\iint\limits_S \operatorname{curl}(\mathbf{F})\,d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$.
   vi. Divergence Theorem: $\iiint\limits_W \operatorname{div} \mathbf{F}\,dV = \iint\limits_S \mathbf{F} \cdot d\mathbf{S}$.
c. Every one of these theorems says that the integral of a derivative of some function over a region equals the "integral" of that function over the boundary of that region.
d. An interesting special case
   i. Given a function $f(x)$, let $\mathbf{F}(x) = f(x)\mathbf{i}$.
   ii. Then $\operatorname{div} \mathbf{F} = f'(x)$.
   iii. Let $[a, b]$ be a region with boundary consisting of the points a and b.
   iv. Orienting the region $[a, b]$ counterclockwise, we have that $\mathbf{i}$ is an outward-pointing normal vector at b and $-\mathbf{i}$ is an outward-pointing normal vector at a.
   v. Thus the total outward flux of $\mathbf{F}$ across the boundary of the region $[a, b]$ is $\mathbf{F}(b) \cdot \mathbf{i} + \mathbf{F}(a) \cdot (-\mathbf{i}) = f(b)\mathbf{i} \cdot \mathbf{i} - f(a)\mathbf{i} \cdot \mathbf{i} = f(b) - f(a)$.
   vi. The Divergence Theorem and Green's Theorem both say that the integral of the divergence of a vector field over a region is equal to the outward flux across the boundary of that region.
   vii. For this region, then, we have $\int_a^b f'(x)\,dx = f(b) - f(a)$.
   viii. In other words, the Fundamental Theorem of Calculus is actually a one-dimensional version of Green's Theorem and the Divergence Theorem.

Homework: Section 17.3 #3, 5, 12, 13, 14, 18, 20, 21.